ResNet18 and XLS-R MIL on the expanded cohort

Saved evaluation complete

Matched patient splits, shared development folds, and saved aggregate results.

Selected XLS-R variant: B · ResNet18 selected epoch: 10 · XLS-R MIL selected epoch: 21

Cohort and shared patient partitions

The cohort combines retained prior NM/AD recordings with the new MCI archive. Prior MCI recordings are excluded. Eligibility uses canonical 06/07 tasks. Distinct repeated takes from the same patient remain eligible; exact duplicate content and unresolved cross-patient identity conflicts must be handled before the manifest is frozen.

Included patients: **234** · Included recordings: **9,080**.

Diagnosis	Patients	Recordings
NM	48	1,839
AD	32	1,065
MCI	154	6,176
Cohort audit		**Count**
Prior MCI patients excluded		18
Prior MCI recordings excluded		711
Repeated task slots with distinct retained takes		30
Recordings retained in those repeated slots		61
Additional distinct takes retained		31
Participants with multiple takes		21

Source and task coverage

Source cohort	Split	Task	Patients	Recordings
Retained prior NM/AD	train	06	56	1,540
Retained prior NM/AD	train	07	55	512
Retained prior NM/AD	val	06	12	335
Retained prior NM/AD	val	07	12	105
Retained prior NM/AD	test	06	12	311
Retained prior NM/AD	test	07	12	101
New MCI archive	train	06	108	3,246
New MCI archive	train	07	107	1,081
New MCI archive	val	06	23	693
New MCI archive	val	07	23	232
New MCI archive	test	06	23	690
New MCI archive	test	07	23	234

Shared frozen splits

Both methods use the same approved patient memberships. All recordings from a person stay together. Prior NM/AD split roles are preserved; new MCI patients are assigned deterministically.

Split	Patients	Recordings	NM patients	AD patients	MCI patients
train	164	6,379	34	22	108
val	35	1,365	7	5	23
test	35	1,336	7	5	23

Method and evaluation protocol

ResNet18 rerun of the s00 method

A randomly initialized single-channel ResNet18 classifies fixed-size 40-band log-mel inputs from mono 16 kHz audio. Training uses random five-second crops and equal expected class mass with equal patient mass within each class. Short clips are padded. Validation/test use up to five deterministic windows per recording; window probabilities are averaged into recording probabilities, then recordings are averaged equally within each patient.

Frozen XLS-R rerun of the s01 method

A pinned frozen multilingual XLS-R encoder processes up to five deterministic five-second waveform windows. Padding masks exclude padded samples and encoder frames. Valid frames are averaged within windows, then windows are averaged equally into recording representations. A trainable LayerNorm → Linear(256) → GELU → Dropout(0.3) projection and diagnosis head use all available recordings in each patient bag. A uses mean MIL; B uses learned attention MIL; C uses the same attention MIL with train-only channel variants. Validation/test always use base representations. Patient cross-entropy uses inverse training-patient class-frequency weights.

Shared selection and once-only final evaluation

Five-fold diagnosis-stratified CV combines the study training and validation patients; all recordings from each patient remain together. ResNet18 and XLS-R A/B/C use identical folds. Fold checkpoints use patient macro F1, with lower unweighted patient loss as the tie-breaker. Mean CV patient macro F1 selects the XLS-R variant; ties use lower mean patient loss, then A → B → C. After configuration is frozen, both final models train on the study training patients and select checkpoints on the study validation patients. Each selected method evaluates the study test patients once. Recording results are supplementary.

Technical source audit

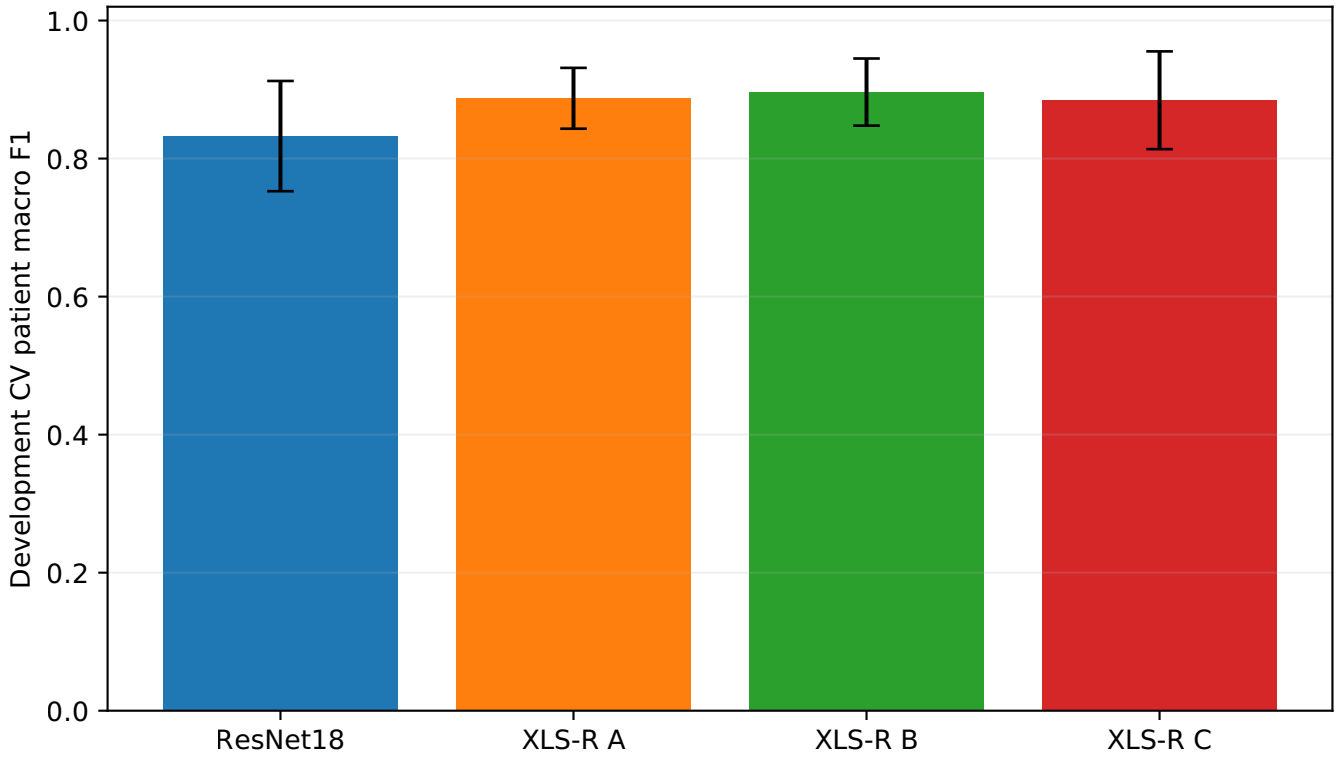
Headers inspected: 9,080 · Header errors: 0. Header inspection includes all splits and performs no model evaluation.

Diagnosis	Split	Recordings	Header errors	Duration min / median / max (s)	Rate / channels / encoding: count
NM	train	1,299	0	1.0073 / 3.4392 / 25.1439	12,000 Hz / 1 ch / PCM_16: 38; 96,000 Hz / 1 ch / PCM_16: 1,260; 96,000 Hz / 1 ch / PCM_24: 1
NM	val	261	0	1.1464 / 3.1874 / 18.0529	96,000 Hz / 1 ch / PCM_16: 261
NM	test	279	0	0.9867 / 3.3746 / 16.9537	96,000 Hz / 1 ch / PCM_16: 279
AD	train	753	0	1.1485 / 3.8305 / 57.9868	96,000 Hz / 1 ch / PCM_16: 752; 96,000 Hz / 1 ch / PCM_24: 1
AD	val	179	0	1.1723 / 4.2250 / 55.3055	96,000 Hz / 1 ch / PCM_16: 177; 96,000 Hz / 1 ch / PCM_24: 2
AD	test	133	0	1.0539 / 3.4954 / 36.1342	96,000 Hz / 1 ch / PCM_16: 132; 96,000 Hz / 1 ch / PCM_24: 1
MCI	train	4,327	0	0.6482 / 3.0543 / 74.8494	16,000 Hz / 1 ch / PCM_16: 4,327
MCI	val	925	0	0.8000 / 3.0556 / 39.1591	16,000 Hz / 1 ch / PCM_16: 925
MCI	test	924	0	0.7904 / 3.0399 / 47.5829	16,000 Hz / 1 ch / PCM_16: 924

These are available technical metadata. They do not establish device/site identities or verify clinical diagnoses.

Shared five-fold development CV

Selected XLS-R variant: **B**. ResNet18 and all XLS-R variants use the same patient folds. Selection uses mean patient macro F1; ties use lower mean patient loss, then A → B → C. Test patients are excluded.



Development CV means; error bars are sample fold SD, not confidence intervals · SVG · PNG

Method	Saved folds	Macro F1 mean ± SD	Balanced accuracy mean ± SD	Macro AUC mean ± SD	Loss mean ± SD
ResNet18	5	0.8326 ± 0.0799	0.8337 ± 0.0826	0.9721 ± 0.0211	0.2026 ± 0.0767
XLS-R A	5	0.8873 ± 0.0441	0.8787 ± 0.0486	0.9773 ± 0.0135	0.2326 ± 0.0713
XLS-R B	5	0.8963 ± 0.0486	0.8994 ± 0.0534	0.9813 ± 0.0084	0.2499 ± 0.0473
XLS-R C	5	0.8845 ± 0.0708	0.8930 ± 0.0661	0.9791 ± 0.0165	0.3285 ± 0.1353

Saved fold checkpoints

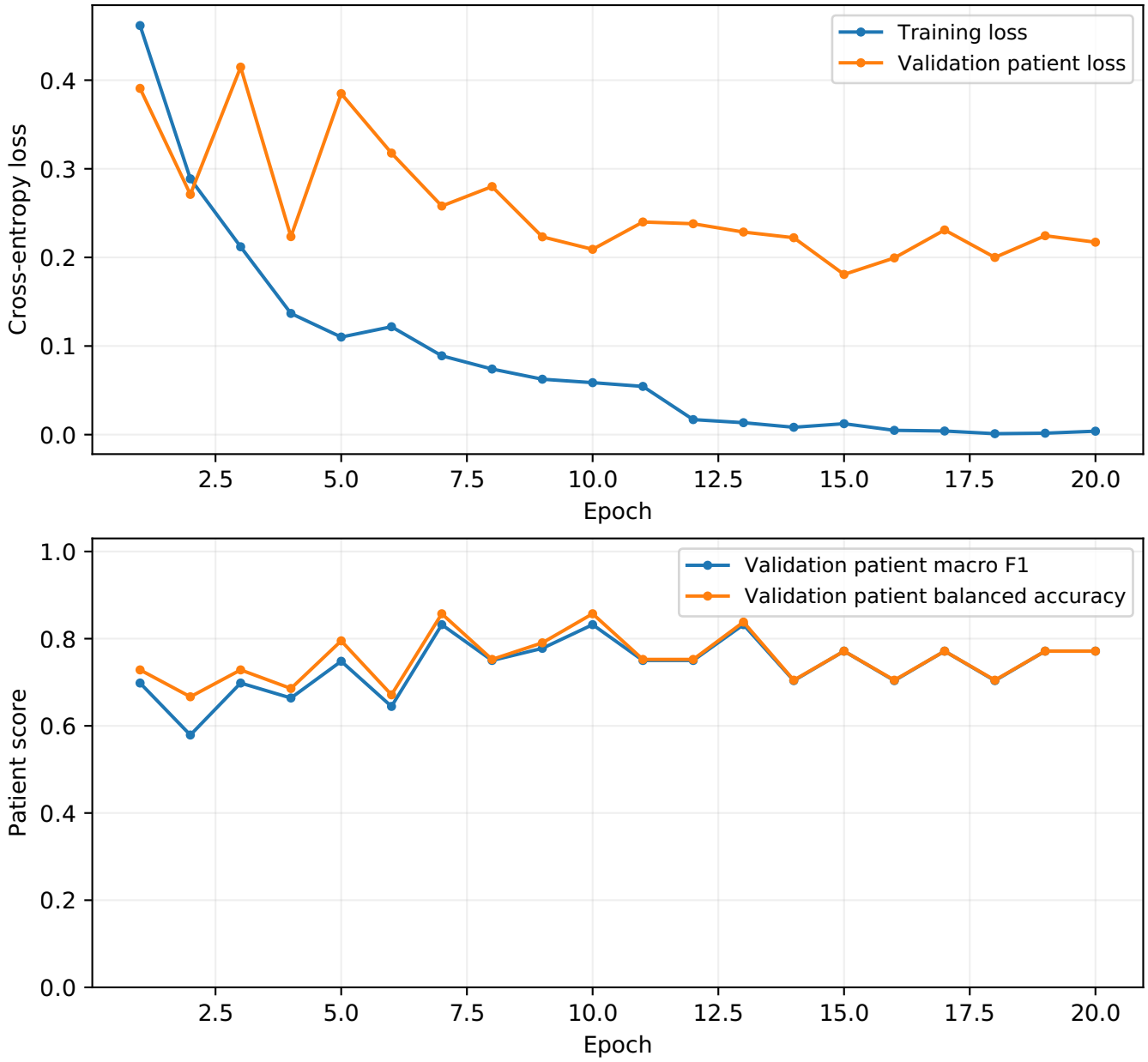
Method	Fold	Best epoch	Patients	Macro F1	Balanced accuracy	Macro AUC	Loss
ResNet18	1	25	40	0.7271	0.7259	0.9380	0.2895
ResNet18	2	4	40	0.9521	0.9583	0.9941	0.1044
ResNet18	3	1	40	0.8235	0.8344	0.9690	0.2551
ResNet18	4	26	40	0.8301	0.8250	0.9808	0.1453
ResNet18	5	7	39	0.8301	0.8250	0.9787	0.2186
XLS-R A	1	15	40	0.8577	0.8296	0.9633	0.3220
XLS-R A	2	19	40	0.9028	0.9028	0.9807	0.2792
XLS-R A	3	31	40	0.9028	0.9028	0.9853	0.2061
XLS-R A	4	70	40	0.9434	0.9333	0.9936	0.1363
XLS-R A	5	31	39	0.8301	0.8250	0.9636	0.2195
XLS-R B	1	37	40	0.8833	0.8667	0.9766	0.2040
XLS-R B	2	15	40	0.9028	0.9028	0.9821	0.2699
XLS-R B	3	20	40	0.8791	0.8900	0.9853	0.3143
XLS-R B	4	25	40	0.9741	0.9877	0.9922	0.2586
XLS-R B	5	41	39	0.8424	0.8500	0.9703	0.2026
XLS-R C	1	24	40	0.8332	0.8761	0.9778	0.3427
XLS-R C	2	8	40	0.8564	0.8611	0.9560	0.5548
XLS-R C	3	43	40	0.9028	0.9028	0.9883	0.2075
XLS-R C	4	21	40	1.0000	1.0000	1.0000	0.2692
XLS-R C	5	17	39	0.8301	0.8250	0.9736	0.2684

Only saved fold results are shown. Missing means remain unavailable; partial folds are never presented as a completed comparison. CV chooses checkpoints and the XLS-R architecture; it is development model selection, not an unbiased nested-CV estimate.

Final-fit learning curves

Each curve describes one method’s final fit. CV fold histories are kept separate.

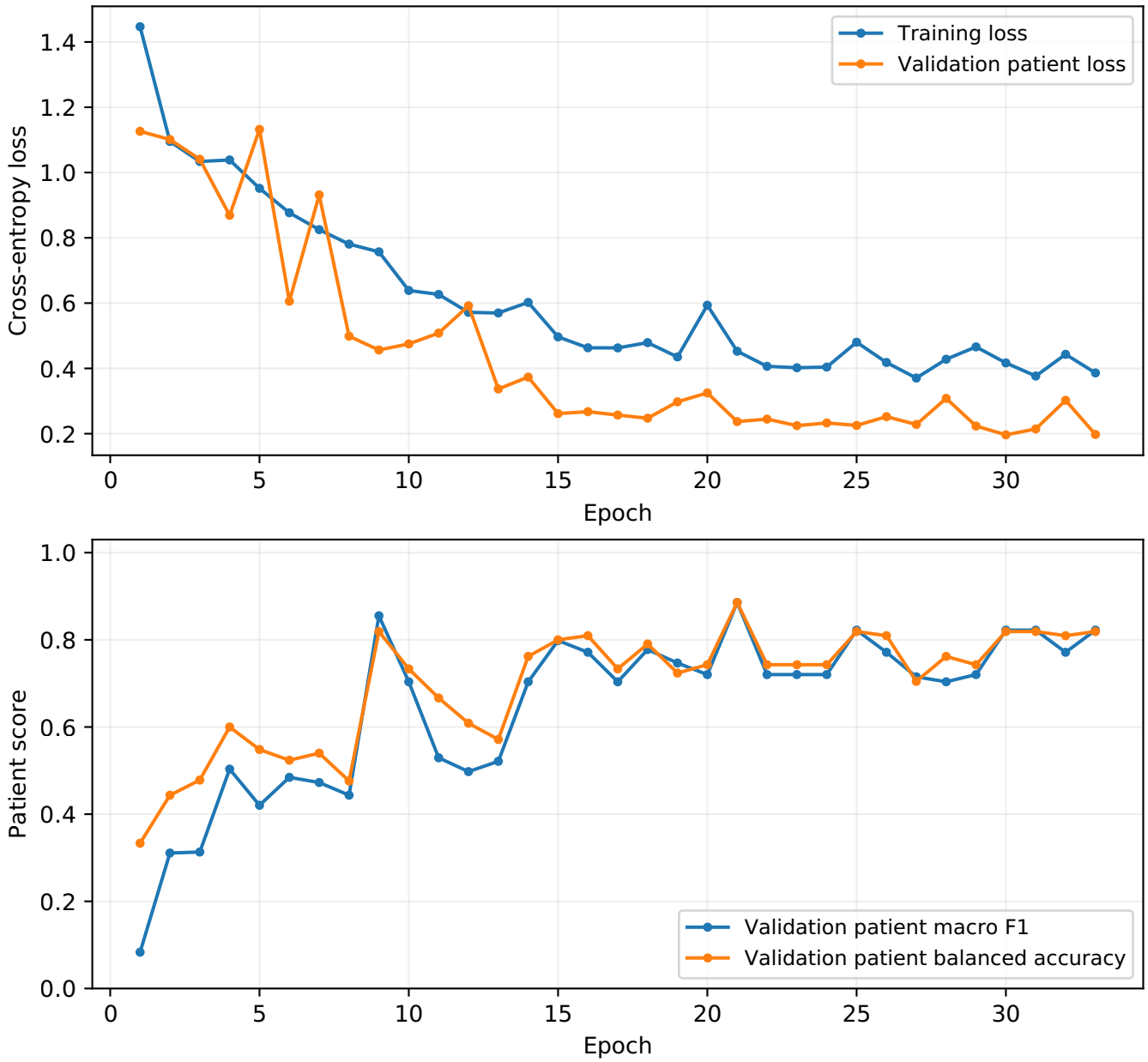
ResNet18



ResNet18: final training and patient-validation history · SVG · PNG

Epoch	Train loss	Patient validation loss	Patient macro F1	Patient balanced accuracy
1	0.4616	0.3907	0.6982	0.7284
2	0.2887	0.2711	0.5789	0.6667
3	0.2120	0.4147	0.6982	0.7284
4	0.1368	0.2234	0.6639	0.6857
5	0.1101	0.3848	0.7481	0.7950
6	0.1218	0.3177	0.6446	0.6712
7	0.0889	0.2580	0.8322	0.8571
8	0.0740	0.2799	0.7500	0.7524
9	0.0625	0.2232	0.7778	0.7905
10	0.0587	0.2091	0.8322	0.8571
11	0.0544	0.2400	0.7500	0.7524
12	0.0170	0.2380	0.7500	0.7524
13	0.0135	0.2286	0.8322	0.8381
14	0.0083	0.2222	0.7037	0.7048
15	0.0123	0.1808	0.7714	0.7714
16	0.0048	0.1994	0.7037	0.7048
17	0.0041	0.2309	0.7714	0.7714
18	0.0011	0.2000	0.7037	0.7048
19	0.0016	0.2245	0.7714	0.7714
20	0.0039	0.2172	0.7714	0.7714

XLS-R MIL



XLS-R MIL: final training and patient-validation history · SVG · PNG

Epoch	Train loss	Patient validation loss	Patient macro F1	Patient balanced accuracy
1	1.4467	1.1263	0.0833	0.3333
2	1.0952	1.1011	0.3108	0.4435
3	1.0338	1.0406	0.3131	0.4783
4	1.0385	0.8689	0.5034	0.6000
5	0.9517	1.1322	0.4204	0.5482
6	0.8768	0.6058	0.4844	0.5238
7	0.8250	0.9315	0.4726	0.5400
8	0.7807	0.4986	0.4435	0.4762
9	0.7571	0.4564	0.8552	0.8190
10	0.6388	0.4750	0.7037	0.7333
11	0.6266	0.5082	0.5294	0.6667
12	0.5717	0.5919	0.4977	0.6087
13	0.5699	0.3371	0.5213	0.5714
14	0.6023	0.3734	0.7037	0.7619
15	0.4966	0.2617	0.7983	0.8000
16	0.4631	0.2675	0.7714	0.8095
17	0.4628	0.2571	0.7037	0.7333
18	0.4790	0.2474	0.7778	0.7905
19	0.4352	0.2976	0.7468	0.7238
20	0.5935	0.3249	0.7203	0.7429
21	0.4526	0.2371	0.8857	0.8857
22	0.4064	0.2447	0.7203	0.7429
23	0.4020	0.2247	0.7203	0.7429
24	0.4041	0.2329	0.7203	0.7429
25	0.4803	0.2254	0.8222	0.8190
26	0.4182	0.2522	0.7714	0.8095
27	0.3705	0.2283	0.7151	0.7048
28	0.4280	0.3081	0.7037	0.7619
29	0.4660	0.2235	0.7203	0.7429
30	0.4169	0.1965	0.8222	0.8190
31	0.3767	0.2145	0.8222	0.8190
32	0.4431	0.3022	0.7714	0.8095
33	0.3862	0.1977	0.8222	0.8190

Matched s02 patient results

These are the current s02 reruns of the ResNet18 method from s00 and XLS-R MIL method from s01. Historical scores from the smaller cohort are not inserted as matched baselines. Each final checkpoint is selected on validation patients before one planned test evaluation per method.

Method	Split	Patients	Macro F1	Balanced accuracy	Accuracy	Macro OvR AUC
ResNet18	validation	35	0.8322	0.8571	0.9143	0.9725
XLS-R MIL	validation	35	0.8857	0.8857	0.9429	0.9765
ResNet18	test	35	0.7203	0.7238	0.8571	0.9686
XLS-R MIL	test	35	0.6571	0.6571	0.8286	0.9166

Missing method results remain pending. Patient identity matching is enforced by the study runner; equal aggregate counts alone cannot establish matching.

ResNet18 · selected checkpoint validation

Patient level · n = 35

Primary patient assessment: ResNet18 averages recording probabilities equally within each patient.

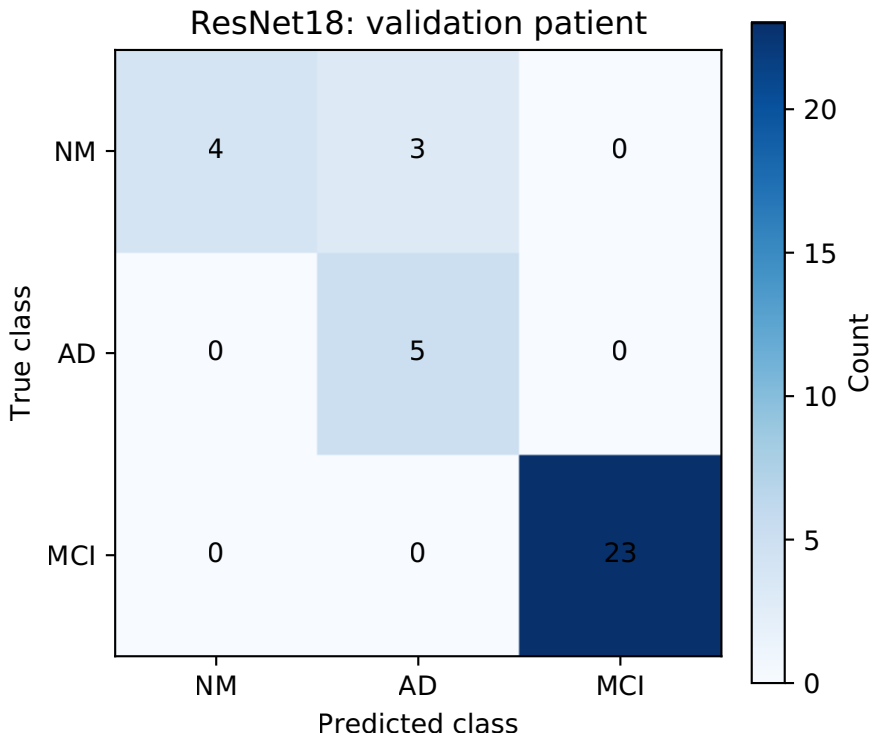
Macro F1	Balanced accuracy	Accuracy	Macro OvR ROC AUC	Loss
83.2%	**85.7%**	**91.4%**	**97.3%**	**0.2091**

Per-class performance

Class	Precision	Recall	F1	OvR ROC AUC	Support
NM	100.0%	57.1%	72.7%	96.4%	7
AD	62.5%	100.0%	76.9%	95.3%	5
MCI	100.0%	100.0%	100.0%	100.0%	23

Confusion matrix

True / predicted	NM	AD	MCI
NM	4	3	0
AD	0	5	0
MCI	0	0	23



ResNet18: validation patient confusion counts · SVG · PNG

Recording level · n = 1,365

Supplementary recording assessment: each recording inherits the patient diagnosis. Correlated clips are not independent patients and never select the architecture or checkpoint.

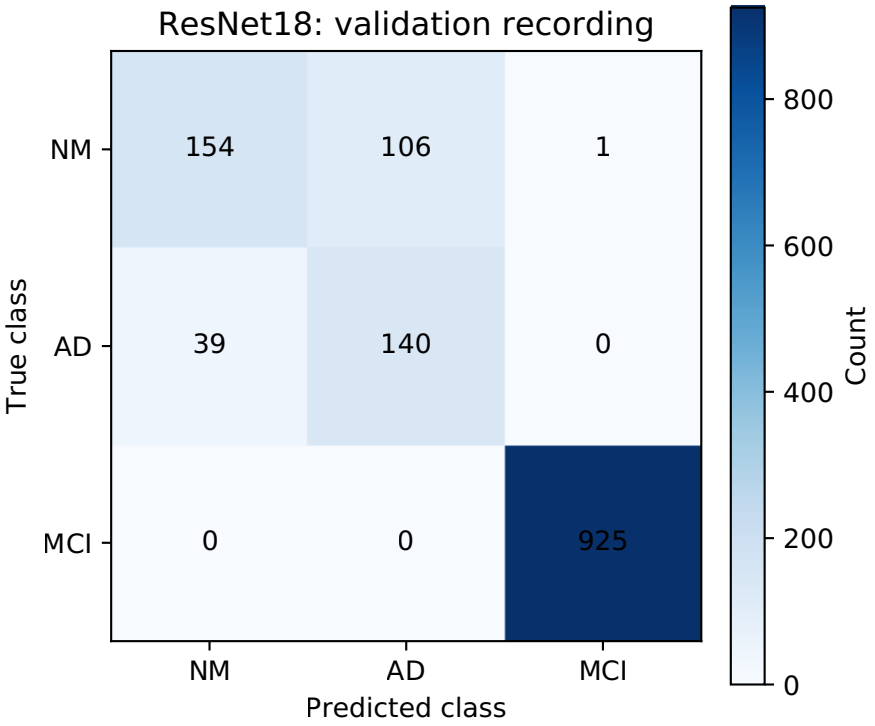
Macro F1	Balanced accuracy	Accuracy	Macro OvR ROC AUC	Loss
77.9%	**79.1%**	**89.3%**	**96.4%**	**0.3543**

Per-class performance

Class	Precision	Recall	F1	OvR ROC AUC	Support
NM	79.8%	59.0%	67.8%	95.3%	261
AD	56.9%	78.2%	65.9%	93.9%	179
MCI	99.9%	100.0%	99.9%	100.0%	925

Confusion matrix

True / predicted	NM	AD	MCI
NM	154	106	1
AD	39	140	0
MCI	0	0	925



ResNet18: validation recording confusion counts · SVG · PNG

AUC uses class probabilities. A class without positive and negative examples has no defined AUC; full three-class macro AUC is unavailable if any class AUC is undefined.

XLS-R MIL · selected checkpoint validation

Patient level · n = 35

Primary patient assessment: one MIL bag prediction uses all available recordings from the patient.

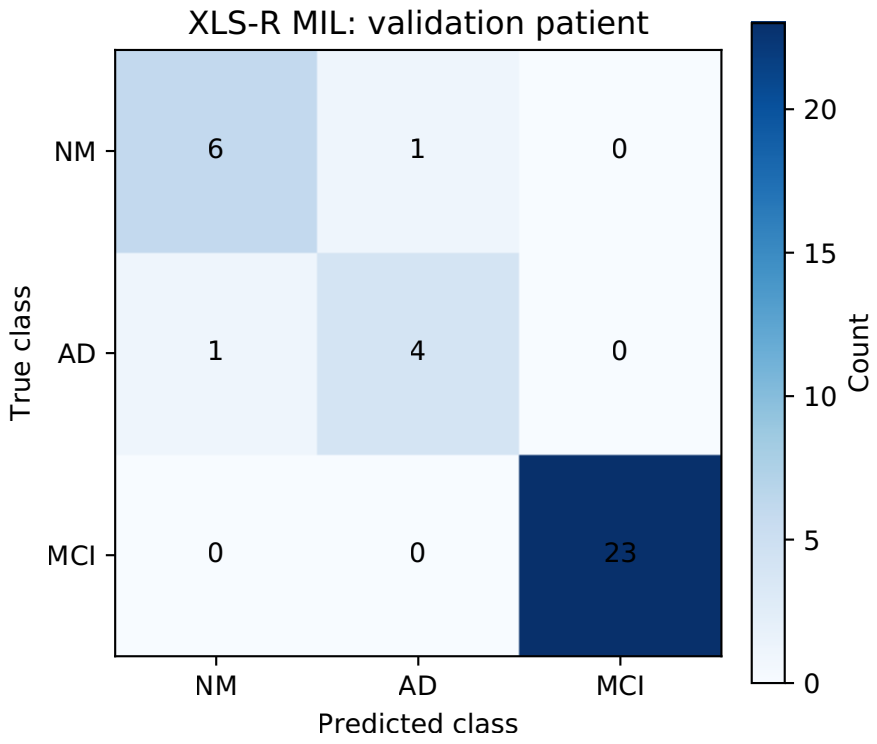
Macro F1	Balanced accuracy	Accuracy	Macro OvR ROC AUC	Loss
88.6%	**88.6%**	**94.3%**	**97.6%**	**0.2371**

Per-class performance

Class	Precision	Recall	F1	OvR ROC AUC	Support
NM	85.7%	85.7%	85.7%	96.9%	7
AD	80.0%	80.0%	80.0%	96.0%	5
MCI	100.0%	100.0%	100.0%	100.0%	23

Confusion matrix

True / predicted	NM	AD	MCI
NM	6	1	0
AD	1	4	0
MCI	0	0	23



XLS-R MIL: validation patient confusion counts · SVG · PNG

Recording level · n = 1,365

Supplementary recording assessment: each recording inherits the patient diagnosis. XLS-R recording predictions use singleton bags. Correlated clips are not independent patients and never select the architecture or checkpoint.

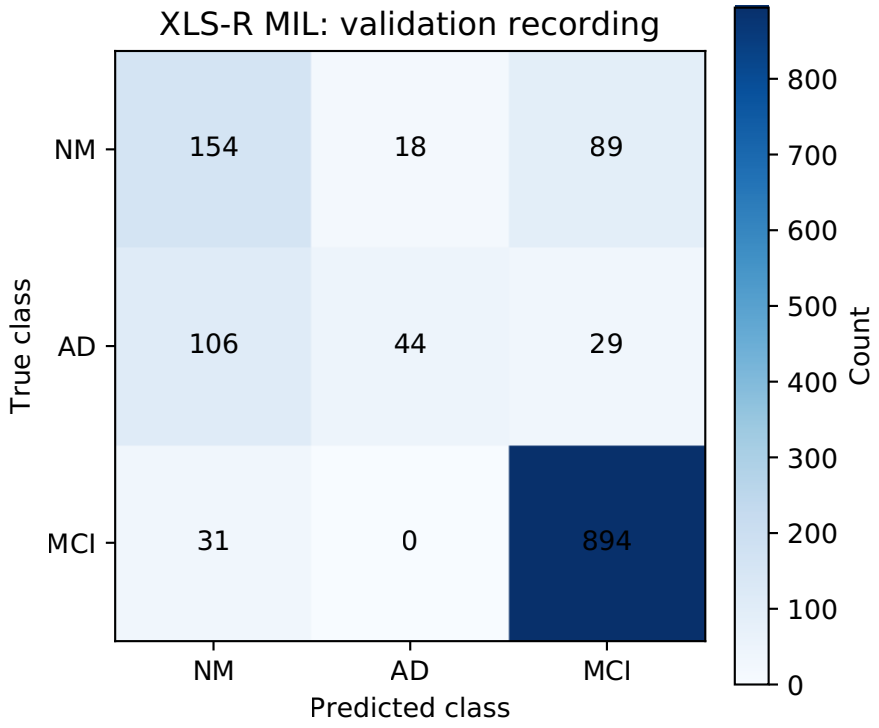
Macro F1	Balanced accuracy	Accuracy	Macro OvR ROC AUC	Loss
61.5%	**60.1%**	**80.0%**	**89.5%**	**0.4923**

Per-class performance

Class	Precision	Recall	F1	OvR ROC AUC	Support
NM	52.9%	59.0%	55.8%	84.8%	261
AD	71.0%	24.6%	36.5%	89.9%	179
MCI	88.3%	96.6%	92.3%	93.7%	925

Confusion matrix

True / predicted	NM	AD	MCI
NM	154	18	89
AD	106	44	29
MCI	31	0	894



XLS-R MIL: validation recording confusion counts · SVG · PNG

AUC uses class probabilities. A class without positive and negative examples has no defined AUC; full three-class macro AUC is unavailable if any class AUC is undefined.

ResNet18 · final test

Patient level · n = 35

Primary patient assessment: ResNet18 averages recording probabilities equally within each patient.

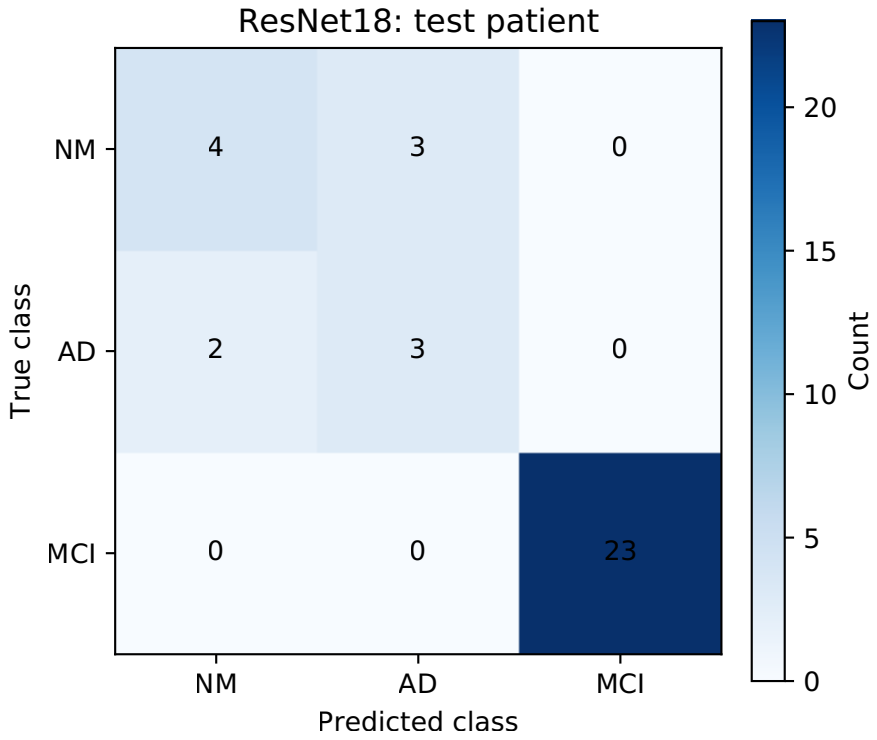
Macro F1	Balanced accuracy	Accuracy	Macro OvR ROC AUC	Loss
72.0%	**72.4%**	**85.7%**	**96.9%**	**0.2271**

Per-class performance

Class	Precision	Recall	F1	OvR ROC AUC	Support
NM	66.7%	57.1%	61.5%	95.9%	7
AD	50.0%	60.0%	54.5%	94.7%	5
MCI	100.0%	100.0%	100.0%	100.0%	23

Confusion matrix

True / predicted	NM	AD	MCI
NM	4	3	0
AD	2	3	0
MCI	0	0	23



ResNet18: test patient confusion counts · SVG · PNG

Recording level · n = 1,336

Supplementary recording assessment: each recording inherits the patient diagnosis. Correlated clips are not independent patients and never select the architecture or checkpoint.

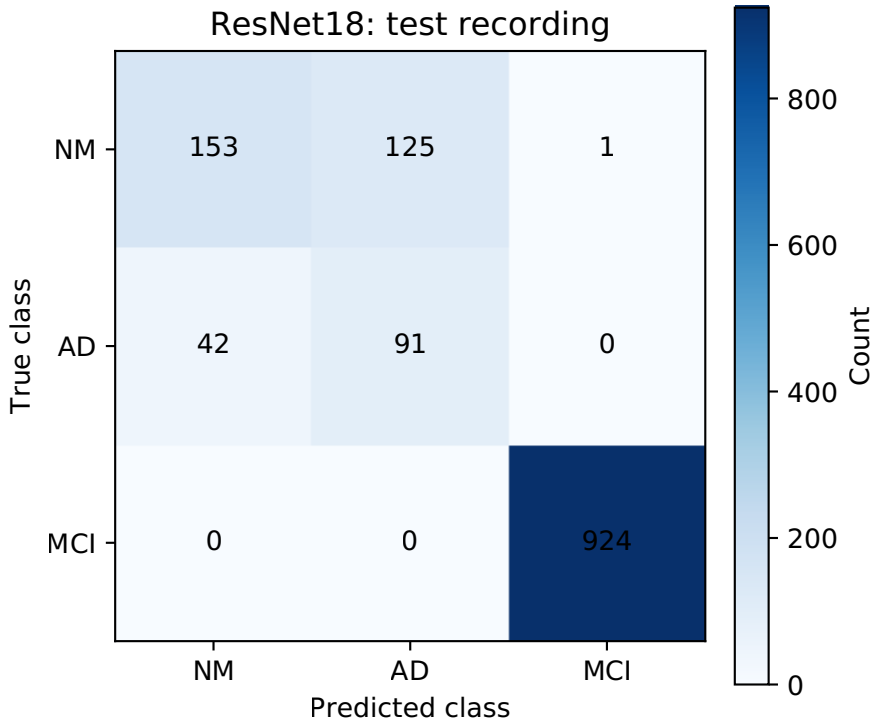
Macro F1	Balanced accuracy	Accuracy	Macro OvR ROC AUC	Loss
72.2%	**74.4%**	**87.4%**	**96.1%**	**0.3785**

Per-class performance

Class	Precision	Recall	F1	OvR ROC AUC	Support
NM	78.5%	54.8%	64.6%	95.8%	279
AD	42.1%	68.4%	52.1%	92.5%	133
MCI	99.9%	100.0%	99.9%	100.0%	924

Confusion matrix

True / predicted	NM	AD	MCI
NM	153	125	1
AD	42	91	0
MCI	0	0	924



ResNet18: test recording confusion counts · SVG · PNG

AUC uses class probabilities. A class without positive and negative examples has no defined AUC; full three-class macro AUC is unavailable if any class AUC is undefined.

XLS-R MIL · final test

Patient level · n = 35

Primary patient assessment: one MIL bag prediction uses all available recordings from the patient.

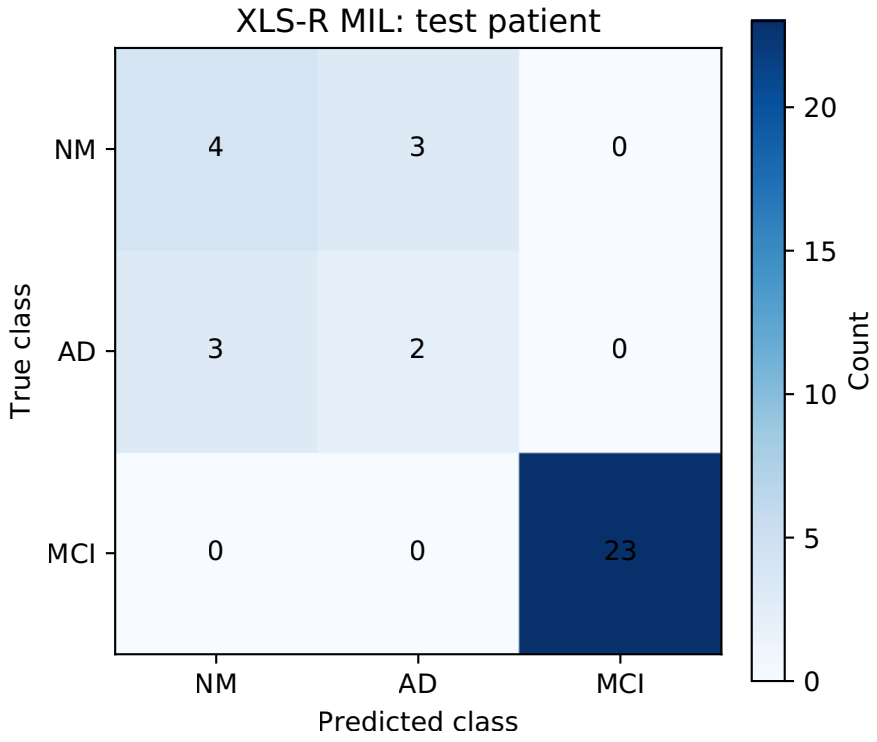
Macro F1	Balanced accuracy	Accuracy	Macro OvR ROC AUC	Loss
65.7%	**65.7%**	**82.9%**	**91.7%**	**0.3046**

Per-class performance

Class	Precision	Recall	F1	OvR ROC AUC	Support
NM	57.1%	57.1%	57.1%	90.3%	7
AD	40.0%	40.0%	40.0%	84.7%	5
MCI	100.0%	100.0%	100.0%	100.0%	23

Confusion matrix

True / predicted	NM	AD	MCI
NM	4	3	0
AD	3	2	0
MCI	0	0	23



XLS-R MIL: test patient confusion counts · SVG · PNG

Recording level · n = 1,336

Supplementary recording assessment: each recording inherits the patient diagnosis. XLS-R recording predictions use singleton bags. Correlated clips are not independent patients and never select the architecture or checkpoint.

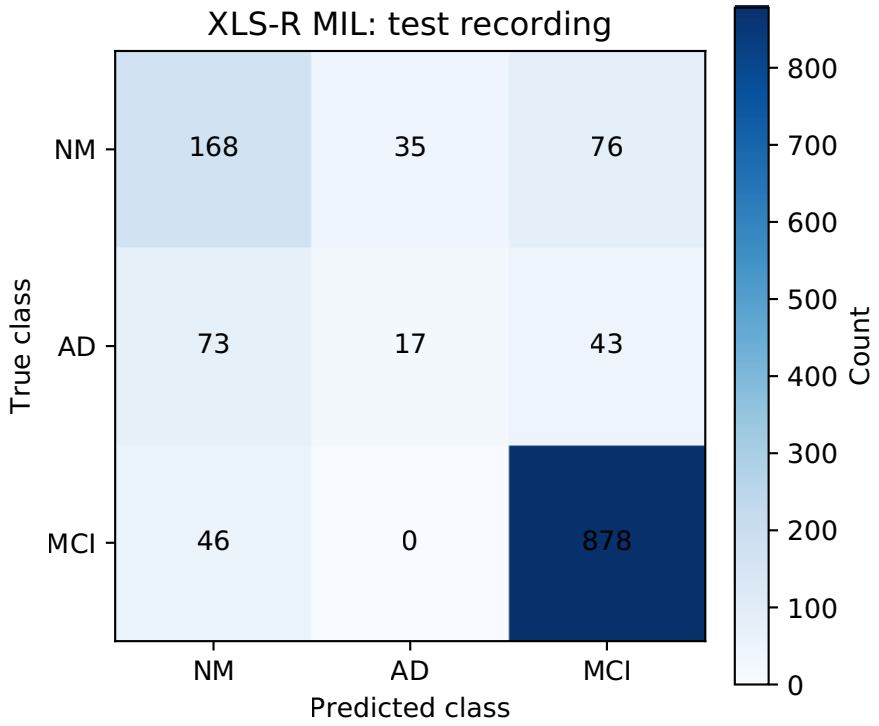
Macro F1	Balanced accuracy	Accuracy	Macro OvR ROC AUC	Loss
56.4%	**56.0%**	**79.6%**	**87.8%**	**0.5096**

Per-class performance

Class	Precision	Recall	F1	OvR ROC AUC	Support
NM	58.5%	60.2%	59.4%	88.0%	279
AD	32.7%	12.8%	18.4%	82.3%	133
MCI	88.1%	95.0%	91.4%	93.2%	924

Confusion matrix

True / predicted	NM	AD	MCI
NM	168	35	76
AD	73	17	43
MCI	46	0	878



XLS-R MIL: test recording confusion counts · SVG · PNG

AUC uses class probabilities. A class without positive and negative examples has no defined AUC; full three-class macro AUC is unavailable if any class AUC is undefined.

Interpretation and practical limitations

The new MCI cohort and prior NM/AD cohort differ in source acquisition. Source and diagnosis are confounded; successful class separation can exploit cohort or channel cues. Resampling and channel transformations do not establish domain invariance. Cross-device or cross-site evidence requires independent devices/sites with diagnosis overlap. Diagnosis must never be treated as a measured device identity.

Prior NM/AD test outcomes were already known from s00/s01, and those patient split roles are retained. This is an expanded-cohort comparison with a partly reused known holdout, not a pristine blinded confirmatory test. Validation and development CV are used for selection and are not independent generalization estimates. Small NM/AD test supports and correlated clips limit precision; inspect AD recall and the full per-class confusion matrices. Strong MCI performance does not by itself demonstrate clinical generalization.

Report patient macro F1, balanced accuracy, AUC and supports together. Class imbalance can make accuracy misleading. Compare only the new matched s02 runs; historical scores from a different cohort do not provide a paired estimate of improvement.

s02 comparison · local figures only · no model inference during report generation